Isabelle Lopez **(818) 964-9755 | izzylopez@ucsb.edu**

EDUCATION

Doctor of Philosophy in Geography — University of California, Santa Barbara
- Spatial Conservation and Landscape Ecology (SCALE) Lab — Sept 2024 - **Present**
- Graduate student affiliate, Center for Spatial Science

Master of Arts in Geography — University of California, Santa Barbara
- Thesis: *Spectrally-derived diversity estimates suggest coastal proximity as a driver of shrubland community composition in a threatened Mediterranean ecosystem* — Sept 2024 - Sept 2026

Bachelor of Science in Wildlife and Conservation Biology — University of New Hampshire
- Minor in Geospatial Analysis — Sept 2019 - May 2023
- Dean's List Recipient: Fall 2021, Spring 2022, Fall 2022, Spring 2023

AWARDS AND FELLOWSHIPS

Pearl Chase Research Grant, UCSB $7,200	2026
Geography Department Block Grant, UCSB $4,000	2026
Graduate Student Scholarship, California Native Plant Society $1,000	2026
Graduate Student Association Grant, UCSB $400	2026
Research Accelerator Award, Schmidt Family Foundation $8,000	2025
Honorable Mention, NSF Graduate Research Fellowship Program	2025
Travel grant, IALE-NA BioLEAPS est. $1,000	2025
Regents in Geography Fellow, UCSB est. $68,000	2024 - 2028
Summer Undergraduate Research Fellow, UNH est. $2,000	2022

PUBLICATIONS

Peer-reviewed:

Lopez, I. S., Fraser, B.T., & Congalton, R.G. (2023). Evaluating the Ability of Unpiloted Aerial Systems to Detect and Monitor Beech Bark Disease in New England Forests. *The Geographical Bulletin*, 64(2). https://digitalcommons.kennesaw.edu/thegeographicalbulletin/vol64/iss2/4

Fraser, B.T., Bunyon, C.L., Reny, S., **Lopez, I. S.**, & Congalton, R.G. (2022). Analysis of Unmanned Aerial System (UAS) Sensor Data for Natural Resource Applications: A Review. *Geographies*, 2(2), Article 2. https://doi.org/10.3390/geographies2020021

Technical Reports:

X. Yan, Y. Zhan, W. Yang, O. Ross, Y. Liu, Z. Li, **I. Lopez**, I. Perez, E. Tao, J. Vogel, D. Cui, L. Song, A. Frazier (2025) Current and Future Habitat Suitability for Special Status Species at the Jack and Laura Dangermond Preserve. UC Santa Barbara: Center for Spatial Science. *EcoEvoRxiv* [Preprint]. https://doi.org/10.32942/X2BH2N

PRESENTATIONS

A.E. Frazier, **I. Lopez**, X. Yan. (2026, October). Leveraging the NASA SHIFT campaign for biodiversity in a threatened Mediterranean ecosystem [Oral presentation]. *USGS Earth Resources Observation and Science (EROS) Center*, Sioux Falls, SD, USA.

Lopez, I, X. Yan, A.E. Frazier. (2026, April). Spectrally-derived diversity estimates indicate topography as a driver of shrubland diversity in a threatened Mediterranean ecosystem [Oral presentation]. *International Association of Landscape Ecologists (IALE) North America,* Athens, GA, USA.

Lopez, I. (2025, September). Shrubland diversity across environmental gradients in a threatened Mediterranean ecosystem [Oral presentation]. *UCSB Department of Geography Seminar*, Santa Barbara, CA, USA.

Lopez, I, X. Yan, A.E. Frazier. (2025, April). Relationships between community assemblage and coastal influence, elevation, and

aspect: Spatial and spectral pattern analysis in a threatened Mediterranean ecosystem [Oral presentation]. *International Association of Landscape Ecologists (IALE) North America,* Raleigh, NC, USA.

Frazier, A.E., D. Cui, L. Song, W. Yang, X. Yan, **I. Lopez**. (2025, April). Species on the Move: planning for future habitat shifts at the Dangermond Preserve under climate change [Oral presentation]. *Point Conception Institute Symposium*, Santa Barbara, CA, USA.

Yan, X, A.E. Frazier, **I. Lopez**. (2025, March). Fire Effects on Plant Diversity: A Hyperspectral Analysis in California's Mediterranean Ecoregion [Oral presentation]. *Association of American Geographers (AAG) Annual Meeting*, Detroit, MI, USA.

Lopez, I, B.T. Fraser, R.G. Congalton. (2023, April). The ability of Unmanned Aerial Systems to detect and monitor Beech Bark Disease [Oral presentation]. *UNH Undergraduate Research Conference,* Durham, NH, USA. **Awarded most outstanding oral presentation.**

Lopez, I, B.T. Fraser, R.G. Congalton. (2023, April). The ability of Unmanned Aerial Systems to detect and monitor Beech Bark Disease [Poster presentation]. *UNH Undergraduate Research Conference,* Durham, NH, USA.

TEACHING EXPERIENCE

Teaching Assistant– University of California, Santa Barbara

GEOG 115A Remote Sensing of the Environment I (Instructor: Joe McFadden)	Fall 2026
GEOG 115C Remote Sensing of the Environment III (Instructor: Zoe Pierrat)	Spring 2026
GEOG 115B Remote Sensing of the Environment II (Instructor: Amy Frazier)	Winter 2026
EARTH 7 Dinosaurs (Instructor: Andy Wyss)	Fall 2025

Guest Lecturer– University of California, Santa Barbara

GEOG 189/289 Close-Range Remote Sensing	Fall 2025

SERVICE AND COMMUNITY ENGAGEMENT

Vice Chair, Graduate Application Materials Assistance Program, UCSB Geography	2026 - Present
Roles Lead, Committee on Committees, UCSB Geography	2026 - Present
Member, California Native Plant Society Channel Islands Chapter Conservation Committee	2026 - Present
Chair, Graduate Application Materials Assistance Program, UCSB Geography	2025 - 2026
Reviewer, Graduate Application Materials Assistance Program, UCSB Geography	2024 - Present
Graduate Student Association Representative, UCSB Geography	2024 - 2025
Undergraduate Representative, Diversity, Equity, and Inclusion Committee, UNH	2021 - 2023

Independent study mentor:

- Adele Wrench, undergraduate, California Polytechnic State University — 2026
- Schuyler Capita, undergraduate, UC Santa Barbara — 2024 - 2026

Geoscience Enrichment and Mentorship for Students by Students (GEMSS) mentor:

- Lily Strange, undergraduate, UC Santa Barbara — 2024 - 2026

Professional memberships:

- American Society for Photogrammetry and Remote Sensing
- Association of American Geographers
- California Native Plant Society
- Southern California Botanists
- International Association of Landscape Ecologists (IALE) North America

RELEVANT RESEARCH & PROFESSIONAL EXPERIENCE

Bren School of Environmental Science & Management— UC Santa Barbara, CA September - November 2025
Environmental Leadership Fellow
- ➢ Served as a graduate student leader in The Nature Conservancy's Environmental Education Program
- ➢ Led groups of high school students on guided educational hikes at the Jack and Laura Dangermond Preserve

NH Fish & Game: Nongame & Endangered Wildlife Division — Concord, NH April - August 2024
Karner Blue Butterfly Biological Technician
- ➢ Worked with a team to captive rear over 150 federally endangered Karner blue butterflies at all life stages in a lab
- ➢ Served as point-person in USFWS-funded Monarch butterfly DNA collection project, coordinated and oversaw humane collection of over 100 Monarch caterpillar individuals across 15 sites throughout New Hampshire

The Nature Conservancy — Newmarket, NH May - December 2023
Seasonal Land Steward & Volunteer Coordinator
- ➢ Served as a field crew leader to two Americorps members, oversaw stewardship work
- ➢ Conducted property inspections for 30 TNC preserves across New Hampshire to identify ecological issues, management problems, and evaluate conservation values

EcoQuest Center for Indigeneity, Ecology and Creativity — Aotearoa, New Zealand September - December 2022
Student Researcher
- ➢ Selected for competitive semester abroad for a hands-on applied field research experience
- ➢ Researched wēta (Orthoptera: Anostostomatidae and Rhaphidophoridae) population dynamics

Basic and Applied Spatial Analysis Lab — University of New Hampshire, NH October 2021 - May 2023
Undergraduate Research Assistant
- ➢ Performed literature review research regarding the use of remote sensing and photogrammetry in grass and shrubland communities as part of a paper published in peer-review academic journal *Geographies*
- ➢ Assisted in UAS (Unmanned Aerial System) flights using both fixed-wing and quadcopter drones

Connors Writing Center — University of New Hampshire, NH September 2021 - May 2023
Undergraduate Writing Assistant
- ➢ Held up to 8 one-on-one writing conferences per week with undergraduate students on projects from any curriculum
- ➢ Assisted students with structure, flow, comprehension, and grammar

Student Conservation Association/U.S. Forest Service — Huron-Manistee National Forests, MI June - August 2021
Botany and Non-Native Invasive Species Intern
- ➢ Identified and removed over 15 invasive species throughout Forest Service properties
- ➢ Assisted in mapping invasive plant communities using ArcGIS Pro

Santa Monica Mountains Fund — Santa Monica Mountains National Recreation Area, CA February - June 2021
Restoration Program Technician
- ➢ Oversaw social media work including short videos as well as weekly posts
- ➢ Assisted in conceptualizing, planning, promoting, and hosting a virtual 500+ people conference associated with milkweed and monarch butterfly conservation

U.S. Department of the Interior — National Park Service Regional Office, Atlanta GA September 2020 - June 2021
Virtual Student Federal Service (VSFS) eIntern
- ➢ Researched the best management practices of invasive species in national parks across the southeast, including 7 national parks and 2 U.S. territories
- ➢ Collaborated with wildlife professionals to create wildlife management protocols

National Park Service — Santa Monica Mountains National Recreation Area, CA November 2020 - January 2021
Social Media Intern
- ➢ Produced, filmed, scripted, and edited short promotional and educational videos for the public
- ➢ Took photographs and videos as well as wrote captions of restoration and nursery work to be used for social media posts

National Park Service — Santa Monica Mountains National Recreation Area, CA May - August 2020
Native Plant Restoration Intern
- ➢ Located, identified, and collected seed from local native plants to use for restoration and preservation efforts

➢ Assisted in general nursery tasks such as transplanting, sowing seed, and watering